Using LaTeX to Type and Reference Mathematical Equations

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Technology: LaTeX using Overleaf

Task: Create professionally formatted mathematical equations and reference them within an academic document.

LaTeX is a document preparation tool that makes it easy to create clean, consistently formatted equations and symbols within written documents. Overleaf is an online LaTeX editor that allows users to collaborate with other users in writing LaTeX code and view the compiled document alongside it. This guide will show you how to typeset mathematical equations and automatically reference them within your writing.

When and Why You Might Use It

Mathematical notation can be difficult to format consistently in a standard word processor. LaTeX uses simple text commands to create clean equations, symbols, subscripts, and superscripts, making it useful whenever mathematical notation appears in a document.

Instructions

1. **Create a project in Overleaf.**

   Open Overleaf and create a blank project. The source code appears on the left side of the editor, while the compiled document appears on the right.

2. **Create an unnumbered equation.**

   Place an equation between `\[` and `\]`:

   ```
   \[
   y = \beta_0 + \beta_1 x + \epsilon
   \]
   ```

   LaTeX automatically centers and formats the equation. Commands beginning with a backslash, such as `\beta` and `\epsilon`, create mathematical symbols, while `_` creates subscripts.

   $$y = \beta_0 + \beta_1 x + \epsilon$$

   For additional mathematical symbols and their corresponding commands, consult the Rice University reference sheet listed in the sources.

3. **Create a numbered equation.**

   Use the `equation` environment when you want LaTeX to automatically assign an equation number:

   ```
   \begin{equation}
       y_t = \beta_0 + \beta_1 x_t + \epsilon_t
       \label{eq:model}
   \end{equation}
   ```

   The `\label{}` command assigns the equation a name that can be used to reference it elsewhere in the document.

4. **Reference the equation in your writing.**

   Instead of manually entering the equation number, use:

   ```
   As shown in Equation \ref{eq:model}, ...
   ```

   LaTeX inserts the correct equation number. If equations are added, removed, or rearranged later, the reference automatically updates.

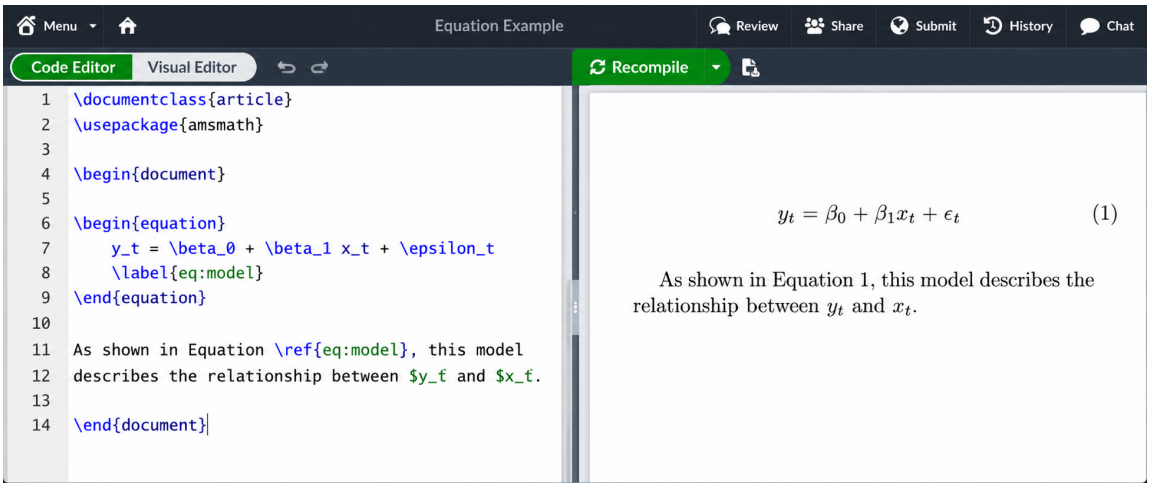


Figure 1: LaTeX source code and its compiled output in Overleaf.

Affordances and Limitations

Affordances: LaTeX consistently formats equations and automatically manages equation numbering and references.

Limitations: LaTeX requires users to learn basic commands, and small syntax errors can prevent a document from compiling correctly.

Access, Usability, and Responsibility

Overleaf is free to use in a web browser, although an account is required to save projects. Paid plans offer additional collaboration features and longer compilation times, but are unnecessary for basic equation formatting. Users should verify that their compiled equations accurately represent their intended meaning.

Sources

Overleaf. "Learn LaTeX in 30 Minutes."
`https://www.overleaf.com/learn/latex/Learn_LaTeX_in_30_minutes`

The LaTeX Project. "LaTeX Documentation."
`https://www.latex-project.org/help/documentation/`

Rice University. "LaTeX Mathematical Symbols."
`https://www.cmor-faculty.rice.edu/~heinken/latex/symbols.pdf`